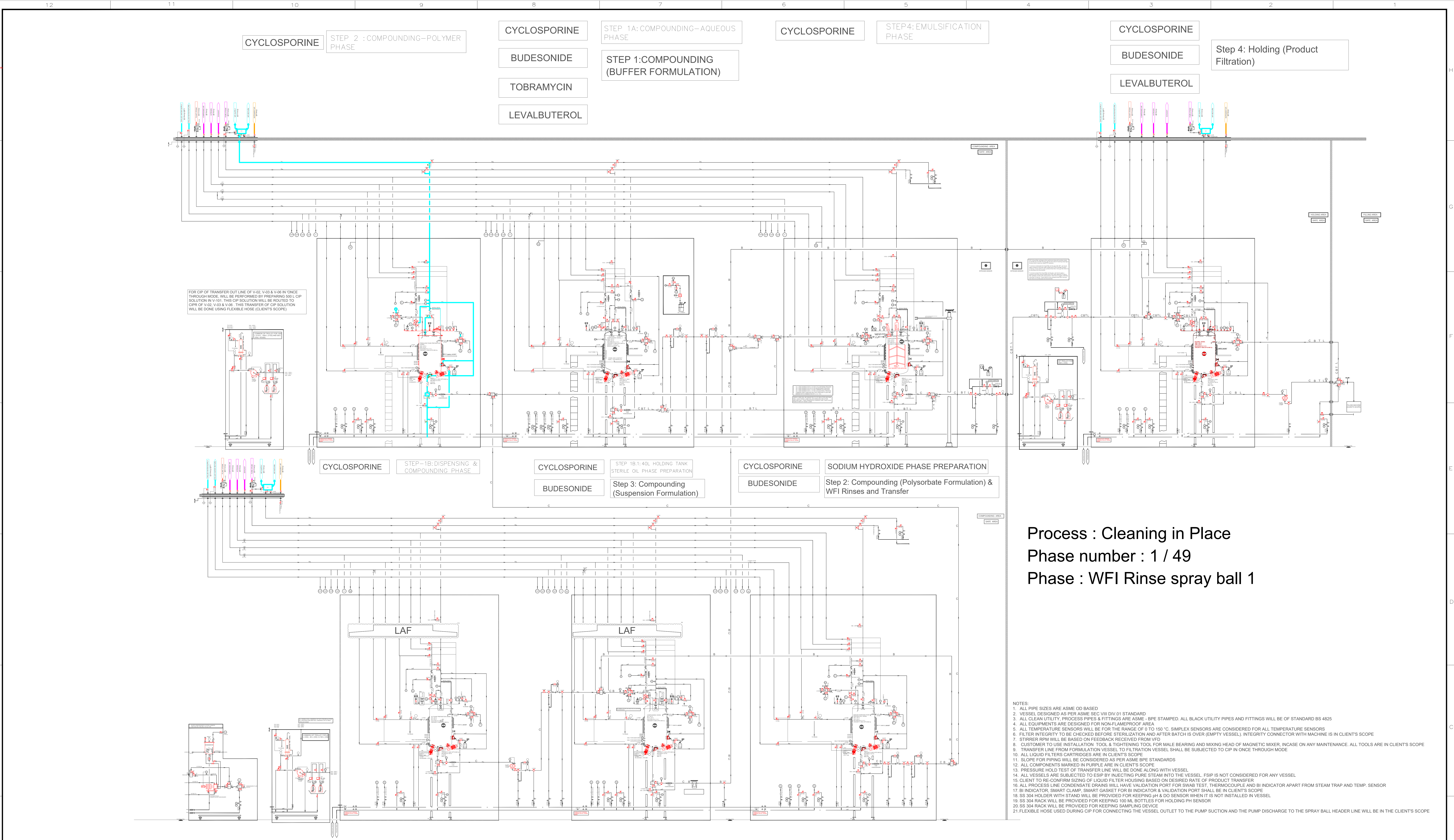


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Process : Cleaning in Place
Phase number : 1 / 49
Phase : WFI Rinse spray ball 1

AREA	COMPOUNDING AREA						HOLDING AREA
PRODUCT	CYCLOSPORINE BUDESONIDE TOBRAMYCN LEVALBUTEROL	CYCLOSPORINE	CYCLOSPORINE	CYCLOSPORINE BUDESONIDE	CYCLOSPORINE BUDESONIDE	CYCLOSPORINE BUDESONIDE	CYCLOSPORINE BUDESONIDE LEVALBUTEROL
VESSLS	V101	V102	V02	V03	V05	V06	V04
GROSS VOL.	1250 L	1250 L	50 L	62.5 L	625 L	125 L	1250 L
WORKING VOL.	1000 L	1000 L	40 L	50 L	500 L	100 L	1000 L
H/D (FOR W.V.)	1.2:1	1.2:1	1.2:1	1.2:1	1.2:1	1.2:1	1.2:1
QTY	1 No.	1 No.	1 No.	1 No.	1 No.	1 No.	1 No.
CONTACT PART	C-22	SS316L	SS316L	SS316L	SS316L	SS316L	SS316L
NON CONTACT PART	SS304	SS304	SS304	SS304	SS304	SS304	SS304
SHELL WORKING PRESSURE	-1 TO 3.0 bar(g)	-1 TO 3.0 bar(g)	-1 TO 3.0 bar(g)	-1 TO 3.0 bar(g)	-1 TO 3.0 bar(g)	-1 TO 3.0 bar(g)	-1 TO 3.0 bar(g)
SHELL DESIGN PRESSURE	-1 TO 4.0 bar(g)	-1 TO 4.0 bar(g)	-1 TO 4.0 bar(g)	-1 TO 4.0 bar(g)	-1 TO 4.0 bar(g)	-1 TO 4.0 bar(g)	-1 TO 4.0 bar(g)
JACKET WORKING PRESSURE	5.2 bar(g)	5.2 bar(g)	5.2 bar(g)	5.2 bar(g)	5.2 bar(g)	5.2 bar(g)	5.2 bar(g)
JACKET DESIGN PRESSURE	-1 TO 3.0 bar(g)	-1 TO 3.0 bar(g)	-1 TO 3.0 bar(g)	-1 TO 3.0 bar(g)	-1 TO 3.0 bar(g)	-1 TO 3.0 bar(g)	-1 TO 3.0 bar(g)
JACKET HYDRO TEST PRESSURE	5.2 bar(g)	5.2 bar(g)	5.2 bar(g)	5.2 bar(g)	5.2 bar(g)	5.2 bar(g)	5.2 bar(g)
SHELL WORKING TEMPERATURE	0° C TO 150° C	0° C TO 150° C	0° C TO 150° C	0° C TO 150° C	0° C TO 150° C	0° C TO 150° C	0° C TO 150° C
SHELL DESIGN TEMPERATURE	0° C TO 180° C	0° C TO 180° C	0° C TO 180° C	0° C TO 180° C	0° C TO 180° C	0° C TO 180° C	0° C TO 180° C
JACKET WORKING TEMPERATURE	0° C TO 150° C	0° C TO 150° C	0° C TO 150° C	0° C TO 150° C	0° C TO 150° C	0° C TO 150° C	0° C TO 150° C
JACKET DESIGN TEMPERATURE	0° C TO 180° C	0° C TO 180° C	0° C TO 180° C	0° C TO 180° C	0° C TO 180° C	0° C TO 180° C	0° C TO 180° C

VESEL SHELL (INTERNAL) : Ra ≤ 0.38 µm, ELECTRO-POLISHED
VESEL SHELL (EXTERNAL) : Ra ≤ 1.2 µm, MATT FINISH
DIAPHRAGM VALVES : PHS will purchase valves with Ra ≤ 0.8 µm. Mechanically polished (FORGED BODY). Electropolishing will be done at PHS Factory to achieve Ra ≤ 0.6 µm
PROCESS & CLEAN UTILITY PIPES : Ras 0.6 µm, ELECTRO-POLISHED
INSTRUMENTS : Ra ≤ 0.8 µm, WHEREVER POSSIBLE
INSULATION (VESSEL) : Rock Wool + PUFF
MOC OF GASKET FOR VESSEL'S BODY FLANGE : VITON
MOC OF GASKET FOR LATERAL VIEW GLASS : VITON
MOC OF O RING FOR INGOLD NOZZLES : VITON
MOC OF GASKET FOR ALL PIPING TC : PTFE
MOC OF GASKET FOR PIPING FLANGE : PTFE

* LINE SECTIONS UP TO 10 IN. (25 CM) IN LENGTH (OR LONGER WITH ADVANCED APPROVAL OF THE OWNER/USER) THAT ARE LEVEL OR HAVE POSITIVE SLOPE LESS THAN THE SLOPE DESIGNATION ARE ACCEPTABLE IF THE SECTION IS FITTING BOUND. *
REFERENCE : ASME BPE 2022 SD 2.4.3

* PHS TO CONSIDER PIPING BATTERY LIMIT ONLY TILL THE FIRST COMPONENT OF THAT PIPE. CONSIDERING FUNCTION ACCESSIBILITY & MAINTAINABILITY OF THAT COMPONENT. PHS TO CONSIDER MINIMUM PIPE & CONNECTION BEFORE THE COMPONENT *

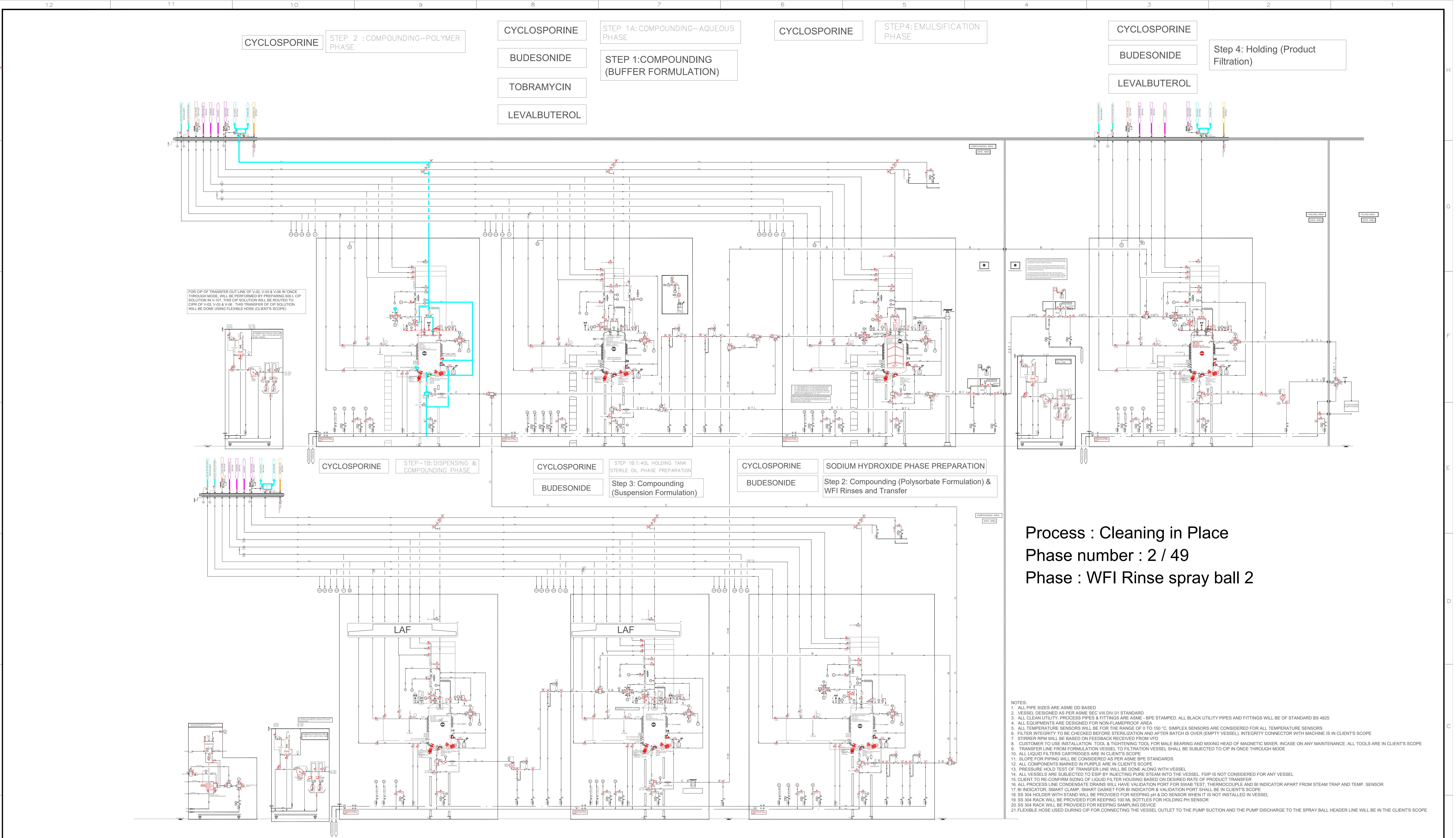
PHS BATTERY LIMIT IS UPTO THE INDIVIDUAL UTILITY PENDANTS.

11	R11	18.07.2025	AS PER CLIENT INPUT -17.07.2025	R.P.	M.M.L.
10	R10	09.07.2025	AS PER CLIENT INPUT -24.06.2025	R.P.	M.M.L.
09	R9	18.06.2025	AS PER CLIENT INPUT -11.06.2025	R.P.	P.B.M.
08	R8	30.05.2025	AS PER CLIENT INPUT -27.05.2025	R.P.	P.B.M.
07	R7	27.05.2025	AS PER INTERNAL REVIEW	S.K.	M.M.L.
06	R6	15.05.2025	AS PER CLIENT INPUT DATED.14.05.2025	S.K.	P.B.M.
05	R5	07.05.2025	AS PER CLIENT INPUT DATED.05.05.2025	S.K.	P.B.M.
04	R4	17.04.2025	AS PER CLIENT INPUT DATED.10.04.2025	R.P.	M.M.L.
03	R3	16.01.2025	AS PER INTERNAL REVIEW	R.P.	A.A.K.
02	R2	09.12.2024	MEETING WITH AMNEAL, DATED : 06.12.2024	R.P.	M.M.L.
01	R1	04.12.2024	MEETING WITH AMNEAL, DATED : 03.12.2024	R.P.	S.K.
SR. NO.	REV. NO.	DATE	DESCRIPTION	REV. BY	DRN. BY

SO. REF. NO.		 Praj HiPurity Systems Limited Unit 1604, DLH Park, S.V Road, Near MTNL Signal, Goregaon (West), Mumbai - 400 104		
CLIENT :-M/s. AMNEAL PHARMACEUTICALS LLC.				
CONSULTANT :-				
DRAWN	M.M.L.	03.12.2024	TITLE :- COMMON P&ID FOR LINE-1	
CHECKED	R.P.	03.12.2024	(CYCLOSPORINE), LINE-2 (BUDESONIDE),	
APPROVED	S.K.	03.12.2024	LINE-3 (TOBRAMYCIN) LINE-4 (LEVALBUTEROL)	
PROJECT NO M-25048	SCALE N.T.S.	SHEET NO 01 OF 01	PROJECTION 	DRAWING NO. M-25048-PID-01-R11

SHEET SIZE A1

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Process : Cleaning in Place
Phase number : 2 / 49
Phase : WFI Rinse spray ball 2

- NOTES:
1. ALL PIPE SIZES ARE ASME OD BASED
 2. VESSEL DESIGNED AS PER ASME SEC VIII DIV 01 STANDARD
 3. ALL CLEAN UTILITY, PROCESS PIPES & FITTINGS ARE ASME - BPE STAMPED. ALL BLACK UTILITY PIPES AND FITTINGS WILL BE OF STANDARD BS 4825
 4. ALL EQUIPMENTS ARE DESIGNED FOR NON-FLAMEPROOF AREA
 5. ALL TEMPERATURE SENSORS WILL BE FOR THE RANGE OF 0 TO 100 °C. SIMPLEX SENSORS ARE CONSIDERED FOR ALL TEMPERATURE SENSORS
 6. FILTER INTEGRITY TO BE CHECKED BEFORE STERILIZATION AND AFTER BATCH IS OVER (EMPTY VESSEL). INTEGRITY CONNECTOR WITH MACHINE IS IN CLIENT'S SCOPE
 7. STIRRER RPM WILL BE BASED ON FEEDBACK RECEIVED FROM VFD
 8. CUSTOMER TO USE INSTALLATION TOOL & TIGHTENING TOOL FOR MALE BEARING AND MIXING HEAD OF MAGNETIC MIXER. IN CASE ON ANY MAINTENANCE, ALL TOOLS ARE IN CLIENT'S SCOPE
 9. TRANSFER LINE FROM FORMULATION VESSEL TO FILTRATION VESSEL SHALL BE SUBJECTED TO CIP IN ONCE THROUGH MODE
 10. ALL LIQUID FILTERS CARTRIDGES ARE IN CLIENT'S SCOPE
 11. SLOPE FOR PIPING WILL BE CONSIDERED AS PER ASME BPE STANDARDS
 12. ALL COMPONENTS MARKED IN PURPLE ARE IN CLIENT'S SCOPE
 13. PRESSURE HOLD TEST OF TRANSFER LINE WILL BE DONE ALONG WITH VESSEL
 14. ALL VESSELS ARE SUBJECTED TO ESB BY INJECTING PURE STEAM INTO THE VESSEL. FSP IS NOT CONSIDERED FOR ANY VESSEL
 15. CLIENT TO RE-CONFIRM SIZING OF LIQUID FILTER HOUSING BASED ON DESIRED RATE OF PRODUCT TRANSFER
 16. ALL PROCESS LINE CONDENSATE DRAINS WILL HAVE VALIDATION PORT FOR SWAB TEST, THERMOCOUPLE AND BI INDICATOR APART FROM STEAM TRAP AND TEMP. SENSOR
 17. BI INDICATOR, SMART CLAMP, SMART GASKET FOR BI INDICATOR & VALIDATION PORT SHALL BE IN CLIENT'S SCOPE
 18. SS 304 HOLDER WITH STAND WILL BE PROVIDED FOR KEEPING pH & DO SENSOR WHEN IT IS NOT INSTALLED IN VESSEL
 19. SS 304 RACK WILL BE PROVIDED FOR KEEPING 100 ML BOTTLES FOR HOLDING PH SENSOR
 20. SS 304 RACK WILL BE PROVIDED FOR KEEPING SAMPLING DEVICE
 21. FLEXIBLE HOSE USED DURING CIP FOR CONNECTING THE VESSEL OUTLET TO THE PUMP SUCTION AND THE PUMP DISCHARGE TO THE SPRAY BALL HEADER LINE WILL BE IN THE CLIENT'S SCOPE

AREA	COMPOUNDING AREA						HOLDING AREA
PRODUCT	CYCLOSPORINE BUDESONIDE TOBRAMYCIN LEVALBUTEROL	CYCLOSPORINE	CYCLOSPORINE	CYCLOSPORINE BUDESONIDE	CYCLOSPORINE BUDESONIDE	CYCLOSPORINE BUDESONIDE LEVALBUTEROL	CYCLOSPORINE BUDESONIDE LEVALBUTEROL
VESSLS	V101	V102	V02	V03	V05	V06	V04
GROSS VOL.	1250 L	1250 L	50 L	62.5 L	625 L	125 L	1250 L
WORKING VOL.	1000 L	1000 L	40 L	50 L	500 L	100 L	1000 L
H/D (FOR W.V.)	1.2:1	1.2:1	1.2:1	1.2:1	1.2:1	1.2:1	1.2:1
QTY	1 No.	1 No.	1 No.	1 No.	1 No.	1 No.	1 No.
CONTACT PART	C-22	SS316L	SS316L	SS316L	SS316L	SS316L	SS316L
NON CONTACT PART	SS304	SS304	SS304	SS304	SS304	SS304	SS304
SHELL WORKING PRESSURE	-1 TO 3.0 bar(g)						
SHELL DESIGN PRESSURE	-1 TO 4.0 bar(g)						
SHELL HYDRO TEST PRESSURE	5.2 bar(g)						
JACKET WORKING PRESSURE	-1 TO 3.0 bar(g)						
JACKET DESIGN PRESSURE	-1 TO 4.0 bar(g)						
JACKET HYDRO TEST PRESSURE	5.2 bar(g)						
SHELL WORKING TEMPERATURE	0° C TO 150° C						
SHELL DESIGN TEMPERATURE	0° C TO 180° C						
JACKET WORKING TEMPERATURE	0° C TO 150° C						
JACKET DESIGN TEMPERATURE	0° C TO 180° C						

VESEL SHELL (INTERNAL) : Ra ≤ 0.38 µm, ELECTRO-POLISHED
VESEL SHELL (EXTERNAL) : Ra ≤ 1.2 µm, MATT FINISH
DIAPHRAGM VALVES : PHS will purchase valves with Ra ≤ 0.8 µm. Mechanically polished (FORGED BODY). Electropolishing will be done at PHS Factory to achieve Ra ≤ 0.6 µm
PROCESS & CLEAN UTILITY PIPES : Ras 0.6 µm, ELECTRO-POLISHED
INSTRUMENTS : Ra ≤ 0.8 µm, WHEREVER POSSIBLE
INSULATION (VESSEL) : Rock Wool + PUFF
MOC OF GASKET FOR VESSEL'S BODY FLANGE : VITON
MOC OF GASKET FOR LATERAL VIEW GLASS : VITON
MOC OF O RING FOR INGOLD NOZZLES : VITON
MOC OF GASKET FOR ALL PIPING TC : PTFE
MOC OF GASKET FOR PIPING FLANGE : PTFE

* LINE SECTIONS UP TO 10 IN. (25 CM) IN LENGTH (OR LONGER WITH ADVANCED APPROVAL OF THE OWNER/USER) THAT ARE LEVEL OR HAVE POSITIVE SLOPE LESS THAN THE SLOPE DESIGNATION ARE ACCEPTABLE IF THE SECTION IS FITTING BOUND. *
REFERENCE : ASME BPE 2022 SD 2.4.3

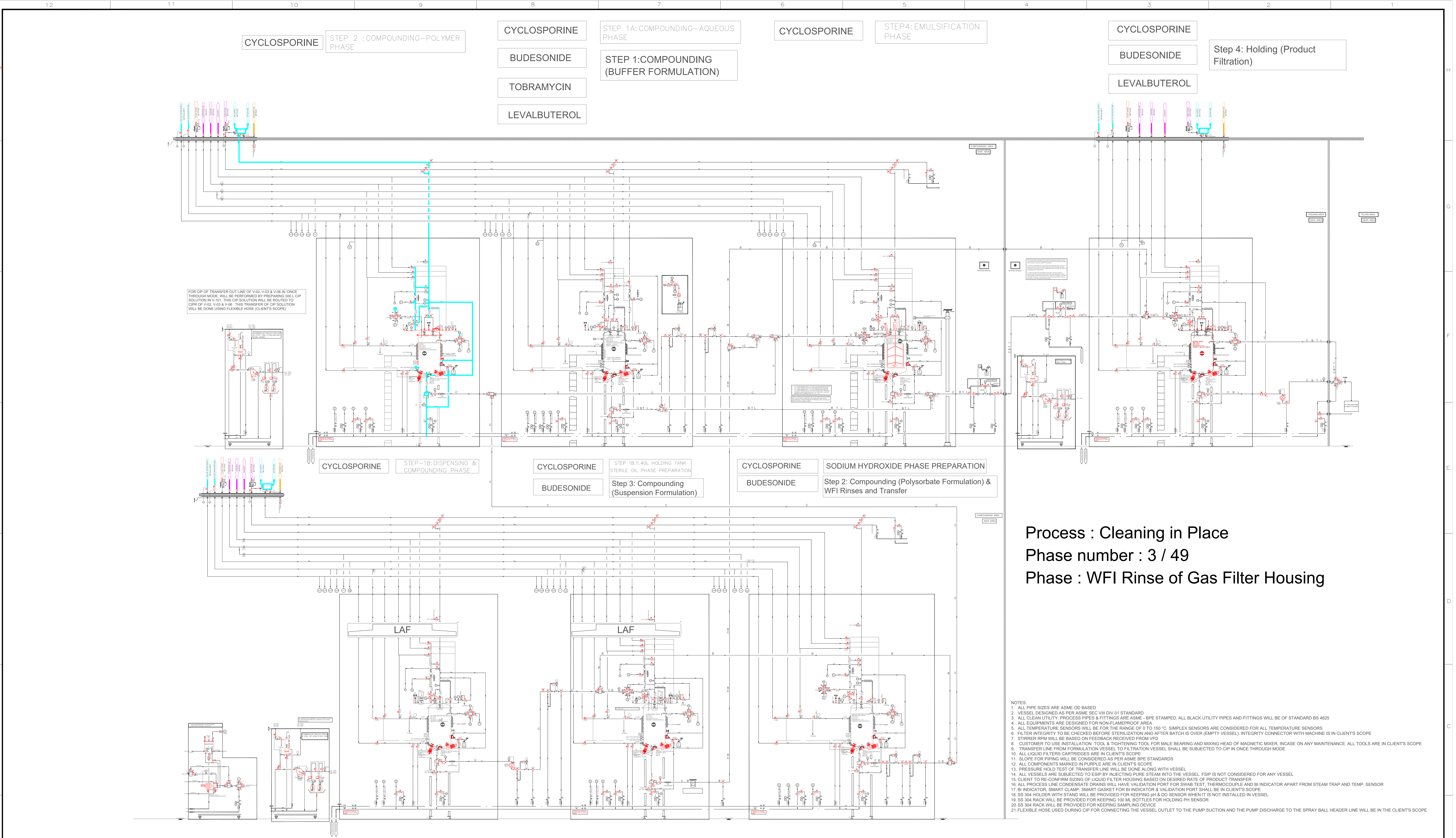
* PHS TO CONSIDER PIPING BATTERY LIMIT ONLY TILL THE FIRST COMPONENT OF THAT PIPE. CONSIDERING FUNCTION ACCESSIBILITY & MAINTAINABILITY OF THAT COMPONENT. PHS TO CONSIDER MINIMUM PIPE & CONNECTION BEFORE THE COMPONENT *

PHS BATTERY LIMIT IS UPTO THE INDIVIDUAL UTILITY PENDANTS.

11	R11	18.07.2025	AS PER CLIENT INPUT -17.07.2025	R.P.	M.M.L.
10	R10	09.07.2025	AS PER CLIENT INPUT -24.06.2025	R.P.	M.M.L.
09	R9	18.06.2025	AS PER CLIENT INPUT -11.06.2025	R.P.	P.B.M.
08	R8	30.05.2025	AS PER CLIENT INPUT -27.05.2025	R.P.	P.B.M.
07	R7	27.05.2025	AS PER INTERNAL REVIEW	S.K.	M.M.L.
06	R6	15.05.2025	AS PER CLIENT INPUT DATED.14.05.2025	S.K.	P.B.M.
05	R5	07.05.2025	AS PER CLIENT INPUT DATED.05.05.2025	S.K.	P.B.M.
04	R4	17.04.2025	AS PER CLIENT INPUT DATED.10.04.2025	R.P.	M.M.L.
03	R3	16.01.2025	AS PER INTERNAL REVIEW	R.P.	A.A.K.
02	R2	09.12.2024	MEETING WITH AMNEAL, DATED : 06.12.2024	R.P.	M.M.L.
01	R1	04.12.2024	MEETING WITH AMNEAL, DATED : 03.12.2024	R.P.	S.K.
SR. NO.	REV. NO.	DATE	DESCRIPTION	REV. BY	DRN. BY

SO. REF. NO.			
 Praj HiPurity Systems Limited Unit 1604, DLH Park, S.V Road, Near MTNL Signal, Goregaon (West), Mumbai - 400 104			
CLIENT :- M/s. AMNEAL PHARMACEUTICALS LLC.			
CONSULTANT :-			
DRAWN	M.M.L.	03.12.2024	TITLE :- COMMON P&ID FOR LINE-1 (CYCLOSPORINE), LINE-2 (BUDESONIDE), LINE-3 (TOBRAMYCIN) LINE-4 (LEVALBUTEROL)
CHECKED	R.P.	03.12.2024	
APPROVED	S.K.	03.12.2024	
PROJECT NO	SCALE	SHEET NO.	PROJECTION
M-25048	N.T.S.	01 OF 01	
DRAWING NO. M-25048-PID-01-R11			

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Process : Cleaning in Place
Phase number : 3 / 49
Phase : WFI Rinse of Gas Filter Housing

AREA	COMPOUNDING AREA						HOLDING AREA
PRODUCT	CYCLOSPORINE BUDESONIDE TOBRAMYCN LEVALBUTEROL	CYCLOSPORINE	CYCLOSPORINE	CYCLOSPORINE BUDESONIDE	CYCLOSPORINE BUDESONIDE	CYCLOSPORINE BUDESONIDE LEVALBUTEROL	CYCLOSPORINE BUDESONIDE LEVALBUTEROL
VESSLS	V101	V102	V02	V03	V05	V06	V04
GROSS VOL.	1250 L	1250 L	50 L	62.5 L	625 L	125 L	1250 L
WORKING VOL.	1000 L	1000 L	40 L	50 L	500 L	100 L	1000 L
H/D (FOR W.V.)	1.2:1	1.2:1	1.2:1	1.2:1	1.2:1	1.2:1	1.2:1
QTY	1 No.	1 No.	1 No.	1 No.	1 No.	1 No.	1 No.
CONTACT PART	C-22	SS316L	SS316L	SS316L	SS316L	SS316L	SS316L
NON CONTACT PART	SS304	SS304	SS304	SS304	SS304	SS304	SS304
SHELL WORKING PRESSURE	-1 TO 3.0 bar(g)	-1 TO 3.0 bar(g)	-1 TO 3.0 bar(g)	-1 TO 3.0 bar(g)	-1 TO 3.0 bar(g)	-1 TO 3.0 bar(g)	-1 TO 3.0 bar(g)
SHELL DESIGN PRESSURE	-1 TO 4.0 bar(g)	-1 TO 4.0 bar(g)	-1 TO 4.0 bar(g)	-1 TO 4.0 bar(g)	-1 TO 4.0 bar(g)	-1 TO 4.0 bar(g)	-1 TO 4.0 bar(g)
JACKET WORKING PRESSURE	5.2 bar(g)	5.2 bar(g)	5.2 bar(g)	5.2 bar(g)	5.2 bar(g)	5.2 bar(g)	5.2 bar(g)
JACKET DESIGN PRESSURE	-1 TO 3.0 bar(g)	-1 TO 3.0 bar(g)	-1 TO 3.0 bar(g)	-1 TO 3.0 bar(g)	-1 TO 3.0 bar(g)	-1 TO 3.0 bar(g)	-1 TO 3.0 bar(g)
JACKET HYDRO TEST PRESSURE	5.2 bar(g)	5.2 bar(g)	5.2 bar(g)	5.2 bar(g)	5.2 bar(g)	5.2 bar(g)	5.2 bar(g)
SHELL WORKING TEMPERATURE	0° C TO 150° C	0° C TO 150° C	0° C TO 150° C	0° C TO 150° C	0° C TO 150° C	0° C TO 150° C	0° C TO 150° C
SHELL DESIGN TEMPERATURE	0° C TO 180° C	0° C TO 180° C	0° C TO 180° C	0° C TO 180° C	0° C TO 180° C	0° C TO 180° C	0° C TO 180° C
JACKET WORKING TEMPERATURE	0° C TO 150° C	0° C TO 150° C	0° C TO 150° C	0° C TO 150° C	0° C TO 150° C	0° C TO 150° C	0° C TO 150° C
JACKET DESIGN TEMPERATURE	0° C TO 180° C	0° C TO 180° C	0° C TO 180° C	0° C TO 180° C	0° C TO 180° C	0° C TO 180° C	0° C TO 180° C

VESEL SHELL (INTERNAL) : Ra ≤ 0.38 µm, ELECTRO-POLISHED
VESEL SHELL (EXTERNAL) : Ra ≤ 1.2 µm, MATT FINISH
DIAPHRAGM VALVES : PHS will purchase valves with Ra ≤ 0.8 µm. Mechanically polished (FORGED BODY). Electropolishing will be done at PHS Factory to achieve Ra ≤ 0.6 µm
PROCESS & CLEAN UTILITY PIPES : Ras 0.6 µm, ELECTRO-POLISHED
INSTRUMENTS : Ra ≤ 0.8 µm, WHEREVER POSSIBLE
INSULATION (VESSEL) : Rock Wool + PUFF
MOC OF GASKET FOR VESSEL'S BODY FLANGE : VITON
MOC OF GASKET FOR LATERAL VIEW GLASS : VITON
MOC OF O RING FOR INGOLD NOZZLES : VITON
MOC OF GASKET FOR ALL PIPING TC : PTFE
MOC OF GASKET FOR PIPING FLANGE : PTFE

* LINE SECTIONS UP TO 10 IN. (25 CM) IN LENGTH (OR LONGER WITH ADVANCED APPROVAL OF THE OWNER/USER) THAT ARE LEVEL OR HAVE POSITIVE SLOPE LESS THAN THE SLOPE DESIGNATION ARE ACCEPTABLE IF THE SECTION IS FITTING BOUND. *
REFERENCE : ASME BPE 2022 SD 2.4.3

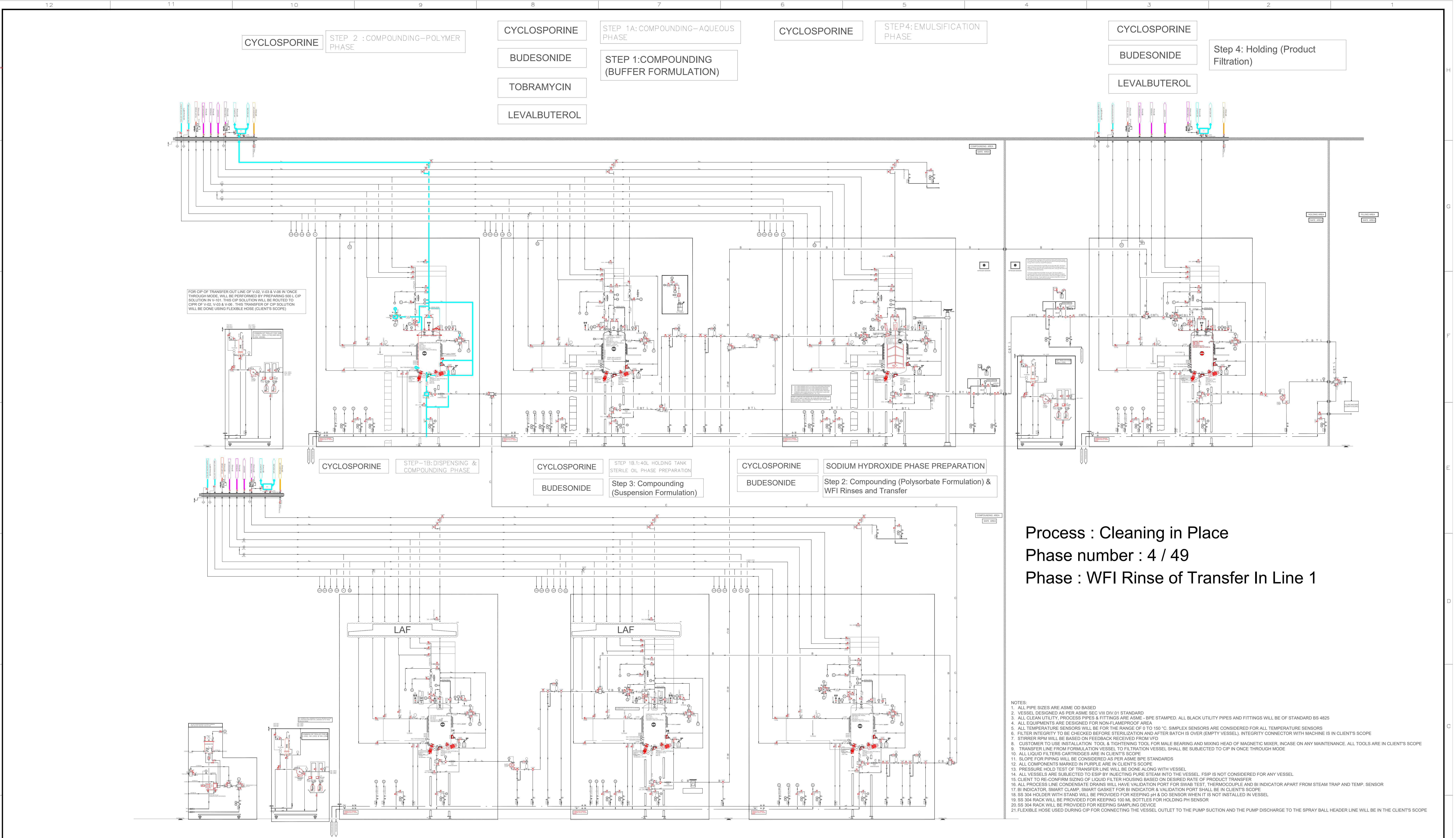
* PHS TO CONSIDER PIPING BATTERY LIMIT ONLY TILL THE FIRST COMPONENT OF THAT PIPE. CONSIDERING FUNCTION ACCESSIBILITY & MAINTAINABILITY OF THAT COMPONENT. PHS TO CONSIDER MINIMUM PIPE & CONNECTION BEFORE THE COMPONENT *

PHS BATTERY LIMIT IS UPTO THE INDIVIDUAL UTILITY PENDANTS.

11	R11	18.07.2025	AS PER CLIENT INPUT -17.07.2025	R.P.	M.M.L.
10	R10	09.07.2025	AS PER CLIENT INPUT -24.06.2025	R.P.	M.M.L.
09	R9	18.06.2025	AS PER CLIENT INPUT -11.06.2025	R.P.	P.B.M.
08	R8	30.05.2025	AS PER CLIENT INPUT -27.05.2025	R.P.	P.B.M.
07	R7	27.05.2025	AS PER INTERNAL REVIEW	S.K.	M.M.L.
06	R6	15.05.2025	AS PER CLIENT INPUT DATED.14.05.2025	S.K.	P.B.M.
05	R5	07.05.2025	AS PER CLIENT INPUT DATED.05.05.2025	S.K.	P.B.M.
04	R4	17.04.2025	AS PER CLIENT INPUT DATED.10.04.2025	R.P.	M.M.L.
03	R3	16.01.2025	AS PER INTERNAL REVIEW	R.P.	A.A.K.
02	R2	09.12.2024	MEETING WITH AMNEAL, DATED : 06.12.2024	R.P.	M.M.L.
01	R1	04.12.2024	MEETING WITH AMNEAL, DATED : 03.12.2024	R.P.	S.K.
SR. NO.	REV. NO.	DATE	DESCRIPTION	REV. BY	DRN. BY

SO. REF. NO.		 Praj HiPurity Systems Limited Unit 1604, DLH Park, S.V Road, Near MTNL Signal, Goregaon (West), Mumbai - 400 104			
CLIENT :-M/s. AMNEAL PHARMACEUTICALS LLC.					
CONSULTANT :-					
DRAWN	M.M.L.	03.12.2024	TITLE :- COMMON P&ID FOR LINE-1 (CYCLOSPORINE), LINE-2 (BUDESONIDE) LINE-3 (TOBRAMYCIN) LINE-4 (LEVALBUTEROL)		
CHECKED	R.P.	03.12.2024			
APPROVED	S.K.	03.12.2024			
PROJECT NO M-25048	SCALE N.T.S.	SHEET NO 01 OF 01	PROJECTION 	DRAWING NO. M-25048-PID-01-R11	

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Process : Cleaning in Place
Phase number : 4 / 49
Phase : WFI Rinse of Transfer In Line 1

AREA	COMPOUNDING AREA						HOLDING AREA
PRODUCT	CYCLOSPORINE BUDESONIDE TOBRAMYCN LEVALBUTEROL	CYCLOSPORINE	CYCLOSPORINE	CYCLOSPORINE BUDESONIDE	CYCLOSPORINE BUDESONIDE	CYCLOSPORINE BUDESONIDE LEVALBUTEROL	CYCLOSPORINE BUDESONIDE LEVALBUTEROL
VESSLS	V101	V102	V02	V03	V05	V06	V04
GROSS VOL.	1250 L	1250 L	50 L	62.5 L	625 L	125 L	1250 L
WORKING VOL.	1000 L	1000 L	40 L	50 L	500 L	100 L	1000 L
H/D (FOR W.V.)	1.2:1	1.2:1	1.2:1	1.2:1	1.2:1	1.2:1	1.2:1
QTY	1 No.	1 No.	1 No.	1 No.	1 No.	1 No.	1 No.
CONTACT PART	C-22	SS316L	SS316L	SS316L	SS316L	SS316L	SS316L
NON CONTACT PART	SS304	SS304	SS304	SS304	SS304	SS304	SS304
SHELL WORKING PRESSURE	-1 TO 3.0 bar(g)						
SHELL DESIGN PRESSURE	-1 TO 4.0 bar(g)						
SHELL HYDRO TEST PRESSURE	5.2 bar(g)						
JACKET WORKING PRESSURE	-1 TO 3.0 bar(g)						
JACKET DESIGN PRESSURE	-1 TO 4.0 bar(g)						
JACKET HYDRO TEST PRESSURE	5.2 bar(g)						
SHELL WORKING TEMPERATURE	0° C TO 150° C						
SHELL DESIGN TEMPERATURE	0° C TO 180° C						
JACKET WORKING TEMPERATURE	0° C TO 150° C						
JACKET DESIGN TEMPERATURE	0° C TO 180° C						

VESEL SHELL (INTERNAL) : Ra ≤ 0.38 µm, ELECTRO-POLISHED
VESEL SHELL (EXTERNAL) : Ra ≤ 1.2 µm, MATT FINISH
DIAPHRAGM VALVES : PHS will purchase valves with Ra ≤ 0.8 µm. Mechanically polished (FORGED BODY). Electropolishing will be done at PHS Factory to achieve Ra ≤ 0.6 µm
PROCESS & CLEAN UTILITY PIPES : Ras 0.6 µm, ELECTRO-POLISHED
INSTRUMENTS : Ra ≤ 0.8 µm, WHEREVER POSSIBLE
INSULATION (VESSEL) : Rock Wool + PUFF
MOC OF GASKET FOR VESSEL'S BODY FLANGE : VITON
MOC OF GASKET FOR LATERAL VIEW GLASS : VITON
MOC OF O RING FOR INGOLD NOZZLES : VITON
MOC OF GASKET FOR ALL PIPING TC : PTFE
MOC OF GASKET FOR PIPING FLANGE : PTFE

11	R11	18.07.2025	AS PER CLIENT INPUT -17.07.2025	R.P.	M.M.L.
10	R10	09.07.2025	AS PER CLIENT INPUT -24.06.2025	R.P.	M.M.L.
09	R9	18.06.2025	AS PER CLIENT INPUT -11.06.2025	R.P.	P.B.M.
08	R8	30.05.2025	AS PER CLIENT INPUT -27.05.2025	R.P.	P.B.M.
07	R7	27.05.2025	AS PER INTERNAL REVIEW	S.K.	M.M.L.
06	R6	15.05.2025	AS PER CLIENT INPUT DATED.14.05.2025	S.K.	P.B.M.
05	R5	07.05.2025	AS PER CLIENT INPUT DATED.05.05.2025	S.K.	P.B.M.
04	R4	17.04.2025	AS PER CLIENT INPUT DATED.10.04.2025	R.P.	M.M.L.
03	R3	16.01.2025	AS PER INTERNAL REVIEW	R.P.	A.A.K.
02	R2	09.12.2024	MEETING WITH AMNEAL, DATED : 06.12.2024	R.P.	M.M.L.
01	R1	04.12.2024	MEETING WITH AMNEAL, DATED : 03.12.2024	R.P.	S.K.

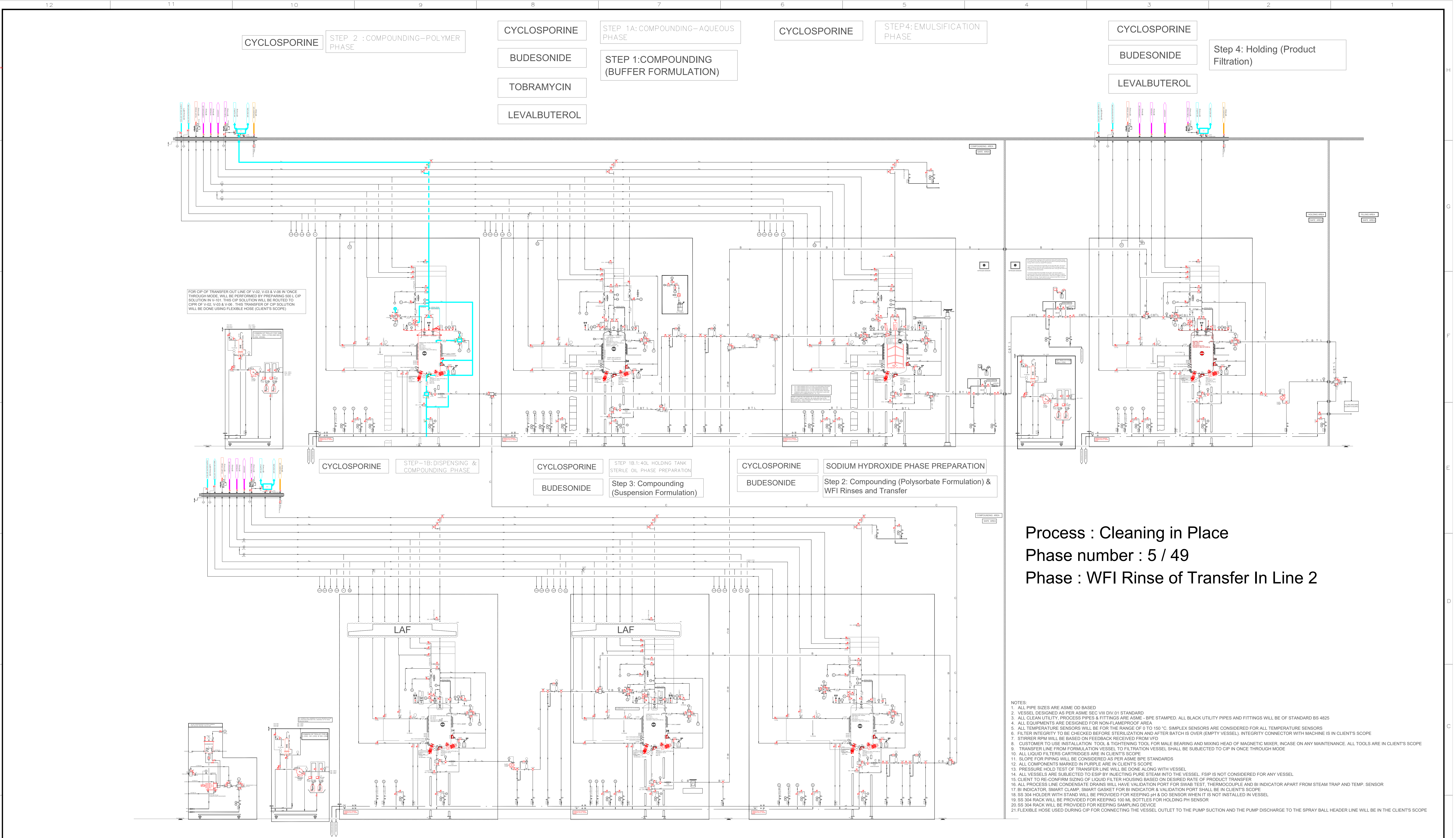
SR. NO.	REV. NO.	DATE	DESCRIPTION	REV. BY	DRN. BY
11	R11	18.07.2025	AS PER CLIENT INPUT -17.07.2025	R.P.	M.M.L.
10	R10	09.07.2025	AS PER CLIENT INPUT -24.06.2025	R.P.	M.M.L.
09	R9	18.06.2025	AS PER CLIENT INPUT -11.06.2025	R.P.	P.B.M.
08	R8	30.05.2025	AS PER CLIENT INPUT -27.05.2025	R.P.	P.B.M.
07	R7	27.05.2025	AS PER INTERNAL REVIEW	S.K.	M.M.L.
06	R6	15.05.2025	AS PER CLIENT INPUT DATED.14.05.2025	S.K.	P.B.M.
05	R5	07.05.2025	AS PER CLIENT INPUT DATED.05.05.2025	S.K.	P.B.M.
04	R4	17.04.2025	AS PER CLIENT INPUT DATED.10.04.2025	R.P.	M.M.L.
03	R3	16.01.2025	AS PER INTERNAL REVIEW	R.P.	A.A.K.
02	R2	09.12.2024	MEETING WITH AMNEAL, DATED : 06.12.2024	R.P.	M.M.L.
01	R1	04.12.2024	MEETING WITH AMNEAL, DATED : 03.12.2024	R.P.	S.K.

UNIT 1604, DLH PARK, S.V. ROAD, NEAR MTNL SIGNAL,
GOREGAON (WEST), MUMBAI - 400 104

SO. REF. NO.

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THIS PRINT IS THE PROPERTY OF Praj HiPurity Systems Limited IT IS TO BE USED ONLY FOR THE PURPOSE FOR WHICH IT WAS LENT AND MUST NOT BE USED ON ANY DETRIMENTAL TO THE INTEREST OF THIS COMPANY AND IT IS SUBJECT TO RETURN ON DEMAND.



Process : Cleaning in Place
Phase number : 5 / 49
Phase : WFI Rinse of Transfer In Line 2

- NOTES:
1. ALL PIPE SIZES ARE ASME OD BASED
 2. VESSEL DESIGNED AS PER ASME SEC VIII DIV 01 STANDARD
 3. ALL CLEAN UTILITY, PROCESS PIPES & FITTINGS ARE ASME - BPE STAMPED. ALL BLACK UTILITY PIPES AND FITTINGS WILL BE OF STANDARD BS 4825
 4. ALL EQUIPMENTS ARE DESIGNED FOR NON-FLAMEPROOF AREA
 5. ALL TEMPERATURE SENSORS WILL BE FOR THE RANGE OF 0 TO 150 °C. SIMPLEX SENSORS ARE CONSIDERED FOR ALL TEMPERATURE SENSORS
 6. FILTER INTEGRITY TO BE CHECKED BEFORE STERILIZATION AND AFTER BATCH IS OVER (EMPTY VESSEL). INTEGRITY CONNECTOR WITH MACHINE IS IN CLIENT'S SCOPE
 7. STIRRER RPM WILL BE BASED ON FEEDBACK RECEIVED FROM VFD
 8. CUSTOMER TO USE INSTALLATION TOOL & TIGHTENING TOOL FOR MALE BEARING AND MIXING HEAD OF MAGNETIC MIXER. INCASE ON ANY MAINTENANCE, ALL TOOLS ARE IN CLIENT'S SCOPE
 9. TRANSFER LINE FROM FORMULATION VESSEL TO FILTRATION VESSEL SHALL BE SUBJECTED TO CIP IN ONCE THROUGH MODE
 10. ALL LIQUID FILTERS CARTRIDGES ARE IN CLIENT'S SCOPE
 11. SLOPE FOR PIPING WILL BE CONSIDERED AS PER ASME BPE STANDARDS
 12. ALL COMPONENTS MARKED IN PURPLE ARE IN CLIENT'S SCOPE
 13. PRESSURE HOLD TEST OF TRANSFER LINE WILL BE DONE ALONG WITH VESSEL
 14. ALL VESSELS ARE SUBJECTED TO ESB BY INJECTING PURE STEAM INTO THE VESSEL. FSP IS NOT CONSIDERED FOR ANY VESSEL
 15. CLIENT TO RE-CONFIRM SIZING OF LIQUID FILTER HOUSING BASED ON DESIRED RATE OF PRODUCT TRANSFER
 16. ALL PROCESS LINE CONDENSATE DRAINS WILL HAVE VALIDATION PORT FOR SWAB TEST, THERMOCOUPLE AND BI INDICATOR APART FROM STEAM TRAP AND TEMP. SENSOR
 17. BI INDICATOR, SMART CLAMP, SMART GASKET FOR BI INDICATOR & VALIDATION PORT SHALL BE IN CLIENT'S SCOPE
 18. SS 304 HOLDER WITH STAND WILL BE PROVIDED FOR KEEPING pH & DO SENSOR WHEN IT IS NOT INSTALLED IN VESSEL
 19. SS 304 RACK WILL BE PROVIDED FOR KEEPING 100 ML BOTTLES FOR HOLDING PH SENSOR
 20. SS 304 RACK WILL BE PROVIDED FOR KEEPING SAMPLING DEVICE
 21. FLEXIBLE HOSE USED DURING CIP FOR CONNECTING THE VESSEL OUTLET TO THE PUMP SUCTION AND THE PUMP DISCHARGE TO THE SPRAY BALL HEADER LINE WILL BE IN THE CLIENT'S SCOPE

AREA	COMPOUNDING AREA						HOLDING AREA
PRODUCT	CYCLOSPORINE BUDESONIDE TOBRAMYCN LEVALBUTEROL	CYCLOSPORINE	CYCLOSPORINE	CYCLOSPORINE BUDESONIDE	CYCLOSPORINE BUDESONIDE	CYCLOSPORINE BUDESONIDE LEVALBUTEROL	CYCLOSPORINE BUDESONIDE LEVALBUTEROL
VESSLS	V101	V102	V02	V03	V05	V06	V04
GROSS VOL.	1250 L	1250 L	50 L	62.5 L	625 L	125 L	1250 L
WORKING VOL.	1000 L	1000 L	40 L	50 L	500 L	100 L	1000 L
H/D (FOR W.V.)	1.2:1	1.2:1	1.2:1	1.2:1	1.2:1	1.2:1	1.2:1
QTY	1 No.	1 No.	1 No.	1 No.	1 No.	1 No.	1 No.
CONTACT PART	C-22	SS316L	SS316L	SS316L	SS316L	SS316L	SS316L
NON CONTACT PART	SS304	SS304	SS304	SS304	SS304	SS304	SS304
SHELL WORKING PRESSURE -1 TO 3.0 bar(g) SHELL DESIGN PRESSURE -1 TO 4.0 bar(g) SHELL HYDRO TEST PRESSURE 5.2 bar(g) JACKET WORKING PRESSURE -1 TO 3.0 bar(g) JACKET DESIGN PRESSURE -1 TO 4.0 bar(g) JACKET HYDRO TEST PRESSURE 5.2 bar(g) SHELL WORKING TEMPERATURE 0° C TO 150° C SHELL DESIGN TEMPERATURE 0° C TO 180° C JACKET WORKING TEMPERATURE 0° C TO 150° C JACKET DESIGN TEMPERATURE 0° C TO 180° C							
VALIDATION PORT SCOPE:- CLIENT'S SCOPE HASTELLOY							

VESSEL SHELL (INTERNAL) : Ra ≤ 0.38 µm, ELECTRO-POLISHED
VESSEL SHELL (EXTERNAL) : Ra ≤ 1.2 µm, MATT FINISH
DIAPHRAGM VALVES : PHS will purchase valves with Ra ≤ 0.8 µm. Mechanically polished (FORGED BODY). Electropolishing will be done at PHS Factory to achieve Ra ≤ 0.6 µm
PROCESS & CLEAN UTILITY PIPES : Ras 0.6 µm, ELECTRO-POLISHED
INSTRUMENTS : Ra ≤ 0.8 µm, WHEREVER POSSIBLE
INSULATION (VESSEL) : Rock Wool + PUFF
MOC OF GASKET FOR VESSEL'S BODY FLANGE : VITON
MOC OF GASKET FOR LATERAL VIEW GLASS : VITON
MOC OF O RING FOR INGOLD NOZZLES : VITON
MOC OF GASKET FOR ALL PIPING TC : PTFE
MOC OF GASKET FOR PIPING FLANGE : PTFE

* LINE SECTIONS UP TO 10 IN. (25 CM) IN LENGTH (OR LONGER WITH ADVANCED APPROVAL OF THE OWNER/USER) THAT ARE LEVEL OR HAVE POSITIVE SLOPE LESS THAN THE SLOPE DESIGNATION ARE ACCEPTABLE IF THE SECTION IS FITTING BOUND. *
REFERENCE : ASME BPE 2022 SD 2.4.3

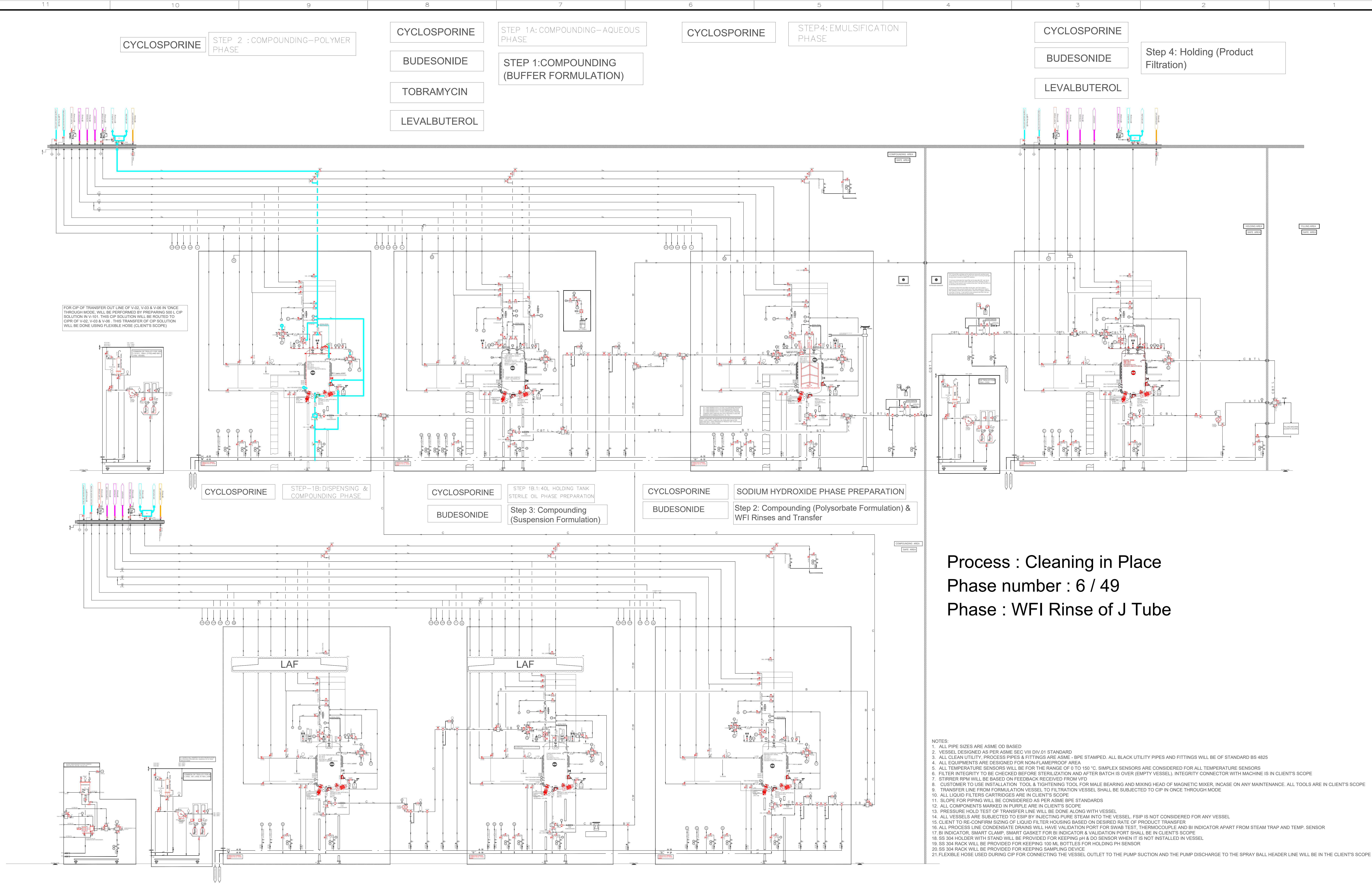
* PHS TO CONSIDER PIPING BATTERY LIMIT ONLY TILL THE FIRST COMPONENT OF THAT PIPE. CONSIDERING FUNCTION ACCESSIBILITY & MAINTAINABILITY OF THAT COMPONENT. PHS TO CONSIDER MINIMUM PIPE & CONNECTION BEFORE THE COMPONENT *

PHS BATTERY LIMIT IS UPTO THE INDIVIDUAL UTILITY PENDANTS.

11	R11	18.07.2025	AS PER CLIENT INPUT -17.07.2025	R.P.	M.M.L.
10	R10	09.07.2025	AS PER CLIENT INPUT -24.06.2025	R.P.	M.M.L.
09	R9	18.06.2025	AS PER CLIENT INPUT -11.06.2025	R.P.	P.B.M.
08	R8	30.05.2025	AS PER CLIENT INPUT -27.05.2025	R.P.	P.B.M.
07	R7	27.05.2025	AS PER INTERNAL REVIEW	S.K.	M.M.L.
06	R6	15.05.2025	AS PER CLIENT INPUT DATED.14.05.2025	S.K.	P.B.M.
05	R5	07.05.2025	AS PER CLIENT INPUT DATED.05.05.2025	S.K.	P.B.M.
04	R4	17.04.2025	AS PER CLIENT INPUT DATED.10.04.2025	R.P.	M.M.L.
03	R3	16.01.2025	AS PER INTERNAL REVIEW	R.P.	A.A.K.
02	R2	09.12.2024	MEETING WITH AMNEAL, DATED : 06.12.2024	R.P.	M.M.L.
01	R1	04.12.2024	MEETING WITH AMNEAL, DATED : 03.12.2024	R.P.	S.K.
SR. NO.	REV. NO.	DATE	DESCRIPTION	REV. BY	DRN. BY

SO. REF. NO.			
Praj HiPurity Systems Limited Unit 1604, DLH Park, S.V Road, Near MTNL Signal, Goregaon (West), Mumbai - 400 104			
CLIENT :- M/s. AMNEAL PHARMACEUTICALS LLC.			
CONSULTANT :-			
DRAWN	M.M.L.	03.12.2024	TITLE :- COMMON P&ID FOR LINE-1 (CYCLOSPORINE), LINE-2 (BUDESONIDE), LINE-3 (TOBRAMYCN) LINE-4 (LEVALBUTEROL)
CHECKED	R.P.	03.12.2024	
APPROVED	S.K.	03.12.2024	
PROJECT NO M-25048	SCALE N.T.S.	SHEET NO. 01 OF 01	PROJECTION
DRAWING NO. M-25048-PID-01-R11			

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Process : Cleaning in Place
Phase number : 6 / 49
Phase : WFI Rinse of J Tube

- NOTES:
1. ALL PIPE SIZES ARE ASME OD BASED
 2. VESSEL DESIGNED AS PER ASME SEC VIII DIV 01 STANDARD
 3. ALL CLEAN UTILITY, PROCESS PIPES & FITTINGS ARE ASME - BPE STAMPED. ALL BLACK UTILITY PIPES AND FITTINGS WILL BE OF STANDARD BS 4825
 4. ALL EQUIPMENTS ARE DESIGNED FOR NON-FLAMEPROOF AREA
 5. ALL TEMPERATURE SENSORS WILL BE FOR THE RANGE OF 0 TO 100 °C. SIMPLEX SENSORS ARE CONSIDERED FOR ALL TEMPERATURE SENSORS
 6. FILTER INTEGRITY TO BE CHECKED BEFORE STERILIZATION AND AFTER BATCH IS OVER (EMPTY VESSEL). INTEGRITY CONNECTOR WITH MACHINE IS IN CLIENT'S SCOPE
 7. STIRRER RPM WILL BE BASED ON FEEDBACK RECEIVED FROM VFD
 8. CUSTOMER TO USE INSTALLATION TOOL & TIGHTENING TOOL FOR MALE BEARING AND MIXING HEAD OF MAGNETIC MIXER. INCASE ON ANY MAINTENANCE, ALL TOOLS ARE IN CLIENT'S SCOPE
 9. TRANSFER LINE FROM FORMULATION VESSEL TO FILTRATION VESSEL SHALL BE SUBJECTED TO CIP IN ONCE THROUGH MODE
 10. ALL LIQUID FILTERS CARTRIDGES ARE IN CLIENT'S SCOPE
 11. SLOPE FOR PIPING WILL BE CONSIDERED AS PER ASME BPE STANDARDS
 12. ALL COMPONENTS MARKED IN PURPLE ARE IN CLIENT'S SCOPE
 13. PRESSURE HOLD TEST OF TRANSFER LINE WILL BE DONE ALONG WITH VESSEL
 14. ALL VESSELS ARE SUBJECTED TO ESB BY INJECTING PURE STEAM INTO THE VESSEL. FSP IS NOT CONSIDERED FOR ANY VESSEL
 15. CLIENT TO RE-CONFIRM SIZING OF LIQUID FILTER HOUSING BASED ON DESIRED RATE OF PRODUCT TRANSFER
 16. ALL PROCESS LINE CONDENSATE DRAINS WILL HAVE VALIDATION PORT FOR SWAB TEST, THERMOCOUPLE AND BI INDICATOR APART FROM STEAM TRAP AND TEMP. SENSOR
 17. BI INDICATOR, SMART CLAMP, SMART GASKET FOR BI INDICATOR & VALIDATION PORT SHALL BE IN CLIENT'S SCOPE
 18. SS 304 HOLDER WITH STAND WILL BE PROVIDED FOR KEEPING pH & DO SENSOR WHEN IT IS NOT INSTALLED IN VESSEL
 19. SS 304 RACK WILL BE PROVIDED FOR KEEPING 100 ML BOTTLES FOR HOLDING PH SENSOR
 20. SS 304 RACK WILL BE PROVIDED FOR KEEPING SAMPLING DEVICE
 21. FLEXIBLE HOSE USED DURING CIP FOR CONNECTING THE VESSEL OUTLET TO THE PUMP SUCTION AND THE PUMP DISCHARGE TO THE SPRAY BALL HEADER LINE WILL BE IN THE CLIENT'S SCOPE

AREA	COMPOUNDING AREA						HOLDING AREA
PRODUCT	CYCLOSPORINE BUDESONIDE TOBRAMYCN LEVALBUTEROL	CYCLOSPORINE	CYCLOSPORINE	CYCLOSPORINE BUDESONIDE	CYCLOSPORINE BUDESONIDE	CYCLOSPORINE BUDESONIDE LEVALBUTEROL	CYCLOSPORINE BUDESONIDE LEVALBUTEROL
VESSLS	V101	V102	V02	V03	V05	V06	V04
GROSS VOL.	1250 L	1250 L	50 L	62.5 L	625 L	125 L	1250 L
WORKING VOL.	1000 L	1000 L	40 L	50 L	500 L	100 L	1000 L
H/D (FOR W.V.)	1.2:1	1.2:1	1.2:1	1.2:1	1.2:1	1.2:1	1.2:1
QTY	1 No.	1 No.	1 No.	1 No.	1 No.	1 No.	1 No.
CONTACT PART	C-22	SS316L	SS316L	SS316L	SS316L	SS316L	SS316L
NON CONTACT PART	SS304	SS304	SS304	SS304	SS304	SS304	SS304
SHELL WORKING PRESSURE	-1 TO 3.0 bar(g)						5.2 bar(g)
SHELL DESIGN PRESSURE	-1 TO 4.0 bar(g)						5.2 bar(g)
JACKET WORKING PRESSURE	-1 TO 3.0 bar(g)						5.2 bar(g)
JACKET DESIGN PRESSURE	-1 TO 4.0 bar(g)						5.2 bar(g)
JACKET HYDRO TEST PRESSURE	5.2 bar(g)						5.2 bar(g)
SHELL WORKING TEMPERATURE	0° C TO 150° C						0° C TO 150° C
SHELL DESIGN TEMPERATURE	0° C TO 180° C						0° C TO 180° C
JACKET WORKING TEMPERATURE	0° C TO 150° C						0° C TO 150° C
JACKET DESIGN TEMPERATURE	0° C TO 180° C						0° C TO 180° C

VESEL SHELL (INTERNAL) : Ra ≤ 0.38 µm, ELECTRO-POLISHED
VESEL SHELL (EXTERNAL) : Ra ≤ 1.2 µm, MATT FINISH
DIAPHRAGM VALVES : PHS will purchase valves with Ra ≤ 0.8 µm. Mechanically polished (FORGED BODY). Electropolishing will be done at PHS Factory to achieve Ra ≤ 0.6 µm
PROCESS & CLEAN UTILITY PIPES : Ras 0.6 µm, ELECTRO-POLISHED
INSTRUMENTS : Ra ≤ 0.8 µm, WHEREVER POSSIBLE
INSULATION (VESSEL) : Rock Wool + PUFF
MOC OF GASKET FOR VESSEL'S BODY FLANGE : VITON
MOC OF GASKET FOR LATERAL VIEW GLASS : VITON
MOC OF O RING FOR INGOLD NOZZLES : VITON
MOC OF GASKET FOR ALL PIPING TC : PTFE
MOC OF GASKET FOR PIPING FLANGE : PTFE

* LINE SECTIONS UP TO 10 IN. (25 CM) IN LENGTH (OR LONGER WITH ADVANCED APPROVAL OF THE OWNER/USER) THAT ARE LEVEL OR HAVE POSITIVE SLOPE LESS THAN THE SLOPE DESIGNATION ARE ACCEPTABLE IF THE SECTION IS FITTING BOUND. *
REFERENCE : ASME BPE 2022 SD 2.4.3

* PHS TO CONSIDER PIPING BATTERY LIMIT ONLY TILL THE FIRST COMPONENT OF THAT PIPE. CONSIDERING FUNCTION ACCESSIBILITY & MAINTAINABILITY OF THAT COMPONENT. PHS TO CONSIDER MINIMUM PIPE & CONNECTION BEFORE THE COMPONENT *

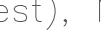
PHS BATTERY LIMIT IS UPTO THE INDIVIDUAL UTILITY PENDANTS.

11	R11	18.07.2025	AS PER CLIENT INPUT -17.07.2025	R.P.	M.M.L.
10	R10	09.07.2025	AS PER CLIENT INPUT -24.06.2025	R.P.	M.M.L.
09	R9	18.06.2025	AS PER CLIENT INPUT -11.06.2025	R.P.	P.B.M.
08	R8	30.05.2025	AS PER CLIENT INPUT -27.05.2025	R.P.	P.B.M.
07	R7	27.05.2025	AS PER INTERNAL REVIEW	S.K.	M.M.L.
06	R6	15.05.2025	AS PER CLIENT INPUT DATED.14.05.2025	S.K.	P.B.M.
05	R5	07.05.2025	AS PER CLIENT INPUT DATED.05.05.2025	S.K.	P.B.M.
04	R4	17.04.2025	AS PER CLIENT INPUT DATED.10.04.2025	R.P.	M.M.L.
03	R3	16.01.2025	AS PER INTERNAL REVIEW	R.P.	A.A.K.
02	R2	09.12.2024	MEETING WITH AMNEAL, DATED : 06.12.2024	R.P.	M.M.L.
01	R1	04.12.2024	MEETING WITH AMNEAL, DATED : 03.12.2024	R.P.	S.K.
SR. NO.	REV. NO.	DATE	DESCRIPTION	REV. BY	DRN. BY

SO. REF. NO.			
Praj HiPurity Systems Limited Unit 1604, DLH Park, S.V Road, Near MTNL Signal, Goregaon (West), Mumbai - 400 104			
CLIENT :- M/s. AMNEAL PHARMACEUTICALS LLC.			
CONSULTANT :-			
DRAWN	M.M.L.	03.12.2024	TITLE :- COMMON P&ID FOR LINE-1 (CYCLOSPORINE), LINE-2 (BUDESONIDE), LINE-3 (TOBRAMYCN) LINE-4 (LEVALBUTEROL)
CHECKED	R.P.	03.12.2024	
APPROVED	S.K.	03.12.2024	
PROJECT NO	SCALE	SHEET NO.	PROJECTION
M-25048	N.T.S.	01 OF 01	
DRAWING NO. M-25048-PID-01-R11			

1. ALL PIPE SIZES ARE ASME OD BASED
2. VESSEL DESIGNED AS PER ASME SEC VIII DIV I STANDARD
3. ALL CLEAN UTILITY, PROCESS PIPES & FITTINGS ARE ASME - BPE STAMPED. ALL BLACK UTILITY PIPES AND FITTINGS WILL BE OF STD STANDARD BS 4825
4. LIQUID FILTERS ARE ORDERED FROM NON-HAZARDOUS AREA
5. ALL TEMPERATURE SENSORS WILL BE FOR THE RANGE OF 0 TO 150 °C. SIMPLEX SENSORS ARE CONSIDERED FOR ALL TEMPERATURE SENSORS
6. FILTER INTEGRITY TO BE CHECKED BEFORE STERILIZATION AND AFTER BATCH IS OVER (EMPTY VESSEL). INTEGRITY CORRELATOR WITH MACHINE IS IN CLIENT'S SCOPE
7. LIQUID FILTER CARTRIDGES ARE IN CLIENT'S SCOPE
8. CUSTOMER TO USE INSTALLATION TOOL & TIGHTENING TOOL FOR MALE BEARING AND MIXING HEAD OF MAGNETIC MECH. IN CASE ON ANY MAINTENANCE. ALL TOOLS ARE IN CLIENT'S SCOPE
9. TRANSFER LINE FROM FORMULATION VESSEL TO FILTRATION VESSEL SHALL BE SUBJECTED TO CP IN C/NCE THROUGH MIDDLE
10. SLOPE FOR PIPING WILL BE CONSIDERED AS PER ASME BPE STANDARDS
11. ALL COMPONENTS MARKED IN PURPLE ARE IN CLIENT'S SCOPE
12. PRESSURE HOLD TEST OF TRANSFER LINE WILL BE DONE ALONG WITH VESSEL
13. ALL VESSELS ARE SUBJECT TO ESP BY INJECTING PURE STEAM INTO THE VESSEL. FSPR IS NOT CONSIDERED FOR ANY CLIENT
14. CLIENT TO BE CONFIRM SIZE OF LIQUID FILL HOUSING BASED ON DESIRED RATE OF PRODUCT TRANSFER
15. ALL PROCESS LINE CONDENSATE DRAINS WILL HAVE VALIDATION PORT FOR SWAB TEST, THERMOCOUPLE AND BI INDICATOR APORT FROM VALVE/DRAIN. TEMP SENSOR
16. BI IN BI INDICATOR GASKET IS IN CLIENT'S SCOPE
17. SS 304 HOLDER WITH STAND WILL BE PROVIDED FOR KEEPING #4 DO SENSOR WHEN IT IS NOT INSTALLED IN VESSEL
18. SS 304 RACK WILL BE PROVIDED FOR KEEPING 100 ML BOTTLES FOR HOLDING SHIP SENSOR
19. BI IN BI INDICATOR GASKET IS IN CLIENT'S SCOPE
20. FLEXIBLE HOSE USED DURING CP FOR CONNECTING THE VESSEL OUTLET TO THE PUMP SUCTION AND THE PUMP DISCHARGE TO THE SPRAY BALL HEADER LINE WILL BE IN THE CLIENT'S SCOPE

VESSEL SHELL (INTERNAL) : Ra $\leq$ 0.38 μ m, ELECTRO-POLISHED
VESSEL SHELL (EXTERNAL) : Ra $\leq$ 1.2 μ m, MATT FINISH
DIAPHRAGM VALVES : PHS will purchase valves with Ra $\leq$ 0.8 μ m, Mechanically polished (FORGED BODY), Electropolishing will be done at PHS Factory to achieve Ra $\leq$ 0.6 μ m
PROCESS & CLEAN UTILITY PIPES : Ra 0.6 μ m, ELECTRO-POLISHED
INSTRUMENTS : Ra $\leq$ 0.8 μ m, WHEREVER POSSIBLE
INSULATION (VESSEL) : Rock Wool + PUFF
MOC OF GASKET FOR VESSEL'S BODY FLANGE : VITON
MOC OF GASKET FOR LATERAL VIEW GLASS : VITON
MOC OF O RING FOR INGOLD NOZZLES : VITON
MOC OF GASKET FOR ALL PIPING TC : PTFE
MOC OF GASKET FOR PIPING FLANGE : PTFE

SO. REF. NO. _____	
 Praj HiPurity Systems Limited Unit 1604, DLH Park, S.V. Road, Near MTNL Signal, Goregaon (West), Mumbai - 400 104	
CLIENT :-M/s. AMNEAL PHARMACEUTICALS LLC.	
CONSULTANT :-	
DRAWN	M.M.L. 03.12.2024
CHECKED	R.P. 03.12.2024
APPROVED	S.K. 03.12.2024
PROJECT NO M-25048	SCALE N.T.S. SHEET NO. 01 OF 01  PROJECTION DRAWING NO. M-25048-PID-01-R11